MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A

VOX MODULE

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope.

This specification sheet provides performance and physical characteristics for the VOX Module used in master Interphone Control panel C-2645D/AIC-14A.

1.2 Military Part Number.

The military part number shall be M22352/7 -1

2. APPLICABLE DOCUMENTS

Refer to 2.2 of MIL-I-22352(AS) for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item Definition. The VOX module shall provide voice activated transmitter key relay activation characteristics as specified herein when used in master Interphone Control panel C-2645D/AIC-14A.

3.1.1 Item Diagram. The VOX module shall provide the functions represented schematically in accordance with the diagram of Figure 1.

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3.1.2 Interface Definition. The VOX Module input and output signals will normally display the characteristics presented in Table I.

TABLE I. Signal characteristics.

Signal Name	Characteristics
Inputs:	
VOX (VOX)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 18 Vac. No dc component normally present.
Key (KEY)	Relay logic: 1) Open 2) Short to ground
VOX Disable (VOX DSABLE)	Relay logic: 1) Open 2) Short to power ground (through a diode)
Power (PWR)	+27.5 ±0.5 Vdc
Outputs:	
Key (KEY)	Relay logic: 1) Open 2) Short to ground
Grounds:	
Power (PWR GND)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS GND)	Signal return. See 3.4.4.3 of MIL-I-22352(AS).

3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

3.2.1.1.1 VOX (VOX). The VOX Module shall have an input for activation of a relay based on the input audio amplitude. The input to the relay closure sensing circuit shall have the following characteristics:

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3.2.1.1.1.1 Frequency. The VOX relay sensing circuitry shall operate in response to inputs over the frequency range from 300 to 7000 Hz.

3.2.1.1.1.2 Amplitude. The VOX input shall accept signals within the specified frequency range which have an amplitude which range from 0 to 18.0 volts. When installed in the C-2645D/AIC-14A, the amplitude of the input will be determined by the level of the audio source, the gain of the microphone circuitry, and by the VOX SENS control internal to the C-2645D/AIC-14A but external to the VOX module.

3.2.1.1.2 Key (KEY). The KEY input shall accept a relay logic signal having two states: 1) open and 2) short to power ground. The KEY input shall enable the VOX circuit to provide a KEY output when required conditions are met.

3.2.1.1.3 VOX Disable (VOX DSABL). The VOX Module shall accept a relay logic input which shall have two states: (1) open and (2) short to ground through a diode. In the open condition the VOX module shall operate normally as described herein. When the VOXOSABL input is grounded, the operation of the VOX relay and associated key functions shall be inhibited and the KEY output shall be a relay logic high.

3.2.1.1.4 Power (PWR). The VOX module shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the amplifier shall not exceed 100 milliamperes.

3.2.1.2 Output Characteristics. The VOX module shall have the following output characteristics:

3.2.1.2.1 Key (KEY). The VOX module shall have a KEY output which shall be a relay logic signal having two states: (1) open and (2) short to ground. The VOX Module KEY output shall present an open circuit for VOX input signals from 300 to 7000 Hz having amplitudes which are below the specified threshold value. Once the specified threshold value has been met the KEY output shall change state to a short circuit to power ground.

Any subsequent decrease in the VOX input amplitude of more than 10 percent below the threshold value shall produce a change in state of the KEY output to an open circuit. The VOX Module, when operated in the circuit of Figure 2, shall produce a short to ground for an input at TP1 which is not more than 0.02 Vac with resistor R1 set to maximum resistance. When resistor R1 is set to minimum resistance, the KEY output shall produce a short to ground when the input amplitude at TP1 is increased to 1.0 Vac. In no case shall the threshold level be greater 0.01 Vac at the VOX input. The attack time shall be a minimum of 50 microseconds and maximum of one millisecond when the input has an amplitude of 5 dB above the threshold value. The release time shall be 750 ±250 milliseconds. VOX switching shall be stable. The VOX output shall not oscillate or chatter when the input is maintained at a constant level near threshold.

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The following definitions shall be used in applying the VOX response criteria.

3.2.1.2.1.1 Threshold. The VOX threshold is defined as the VOX input amplitude level required at the VOX input to energize the key relay, producing the short to ground output condition at the KEY output.

3.2.1.2.1.2 Attack Time. Attack time is defined as the time required to close the relay following application of a 1,000 Hz signal to the VOX input which meets or exceeds the sensing circuit threshold.

3.2.1.2.1.3 Release Time. Release time is defined as the time required to open the output circuit after the removal of the 1,000 Hz signal at the VOX input.

3.2.2 Physical Characteristics

3.2.2.1 Dimensions. The VOX Module shall consist of a single encapsulated module having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The amplifier shall weigh no more than 2.0 ounces.

3.3 Design and Construction. The design and construction of the amplifier shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

4.1 Responsibility for Tests and Inspections. See MIL-I-22352(AS).

4.2 Test Equipment. See 4.3.3.1 MIL-I-22352(AS).

4.3 Test Methods.

4.3.1 Examination of Product. See 4.3.3.2.1 MIL-I-22352(AS).

4.3.2 Operational Test. See 4.3.3.2.2 MIL-I-22352(AS).

4.3.2.1 Threshold. Verify the threshold requirements of 3.2.1.2.1 herein.

4.3.2.2 Attack Time. Verify the attack time requirements of 3.2.1.2.1 herein.

4.3.2.3 Release Time. Verify the release time requirements of 3.2.1.2.1 herein.

Preparing Activity:
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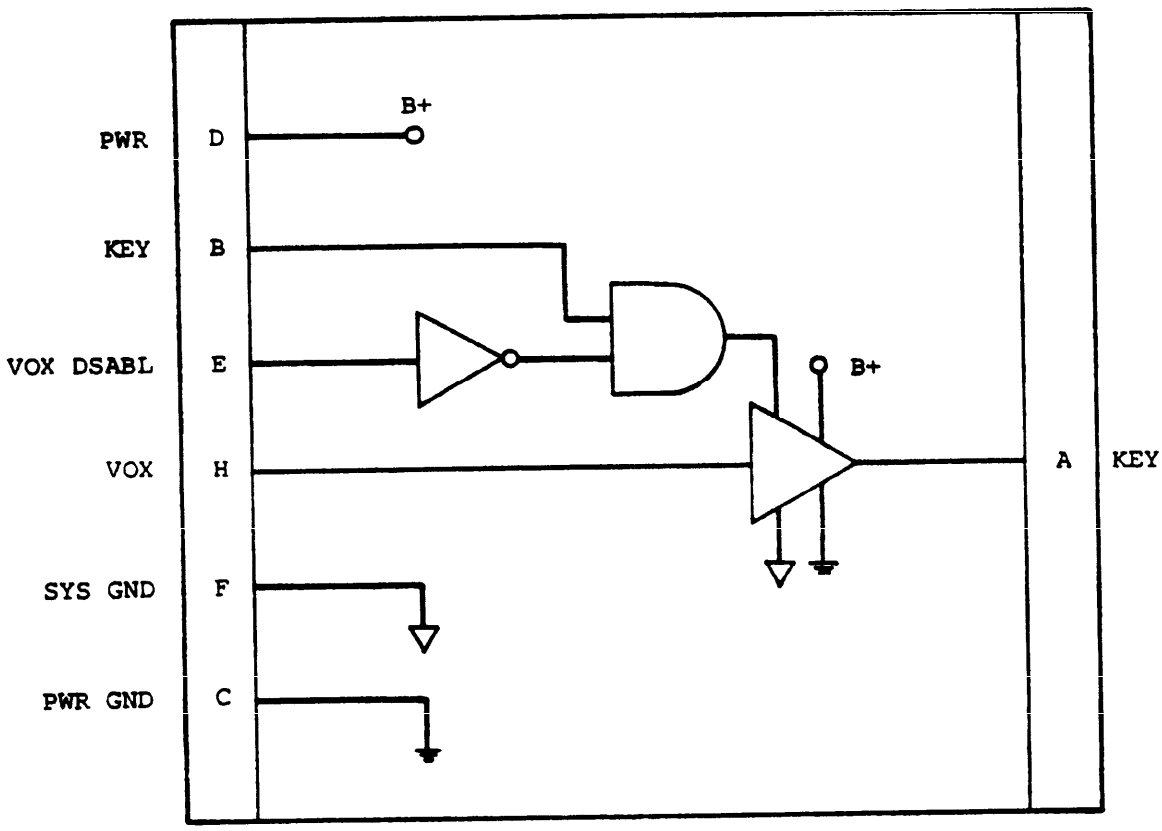


FIGURE 1. VOX module functional diagram.

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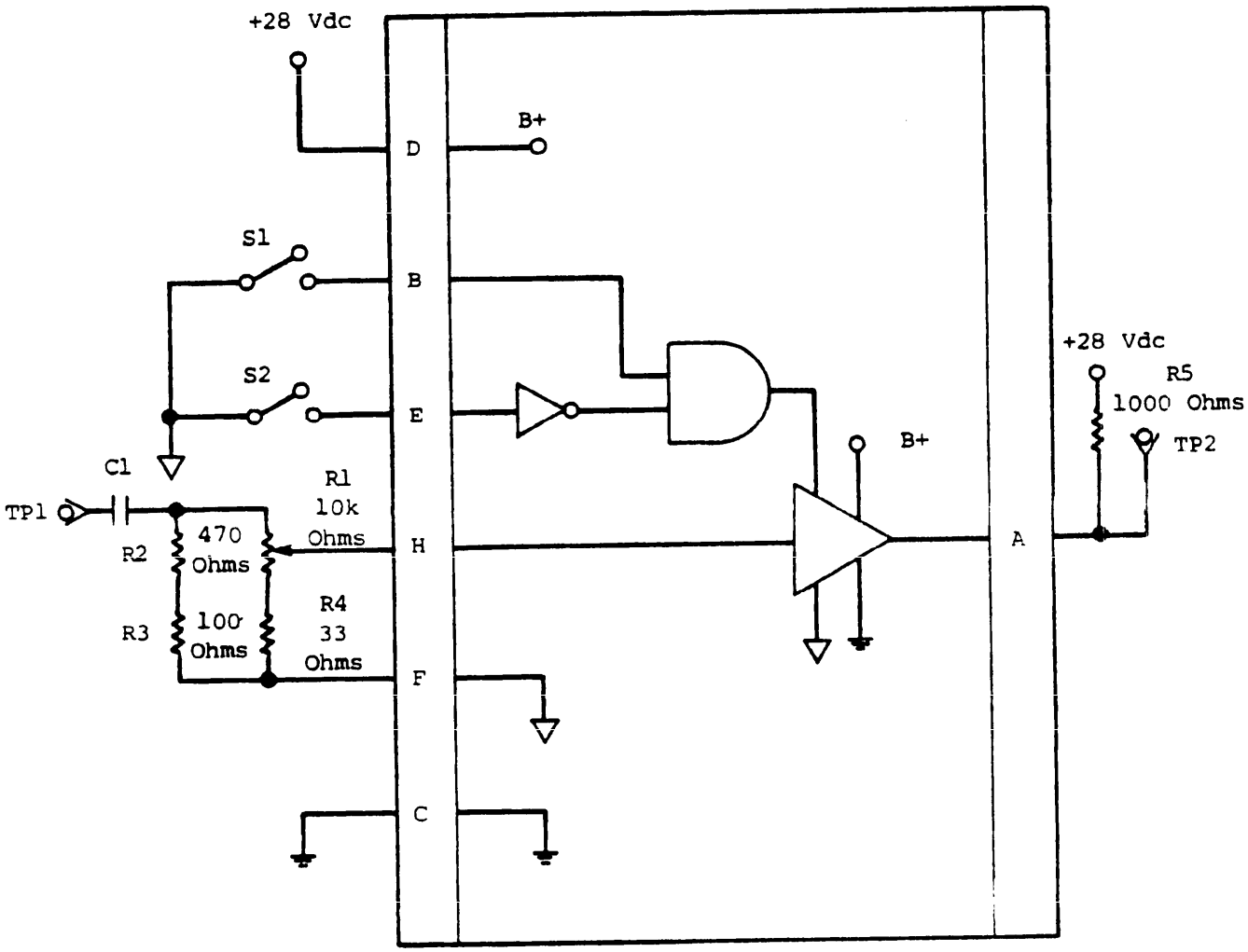
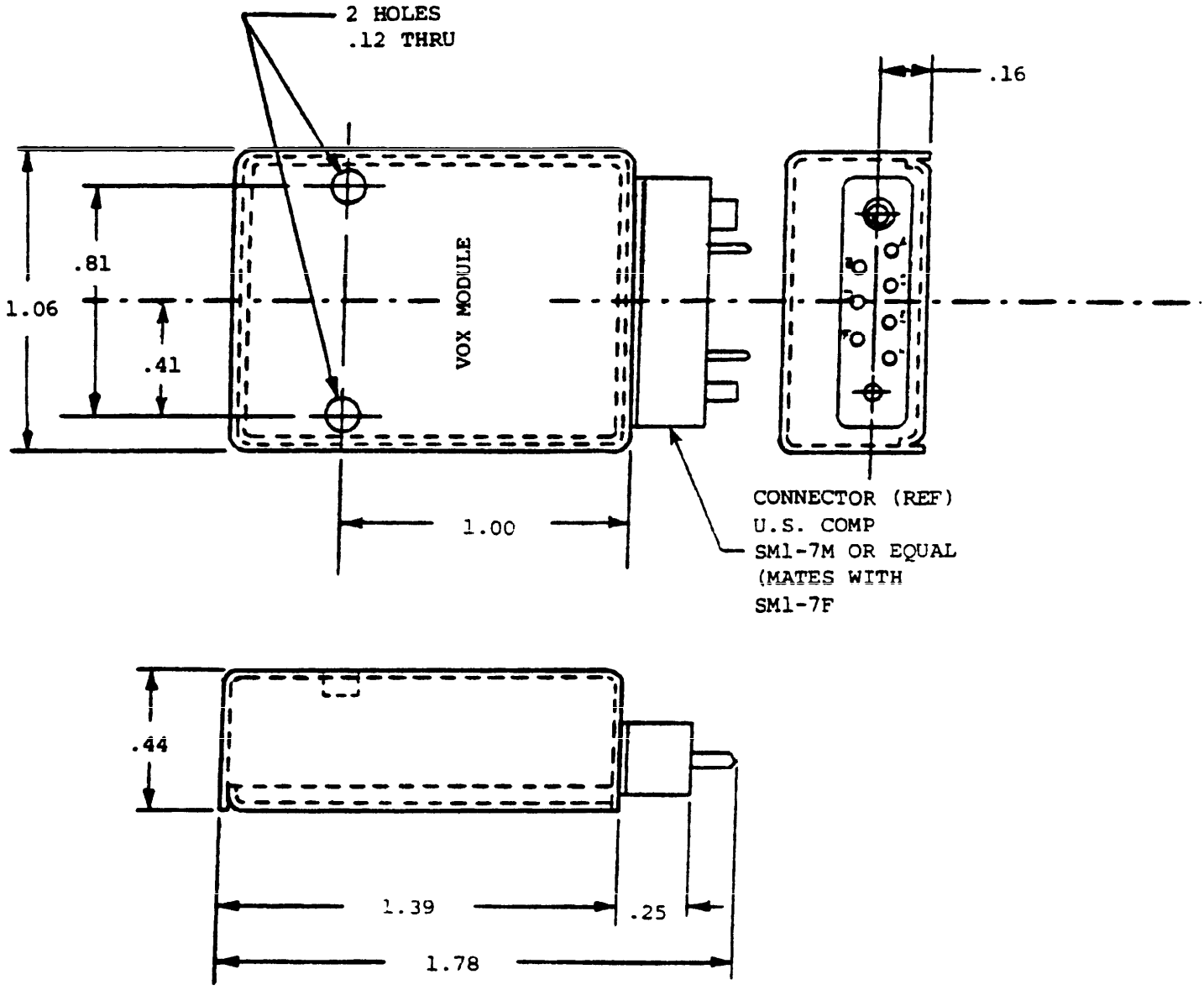


FIGURE 2. VOX module test set-up.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 3. VOX module outline drawing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/7(AS)	INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A VOX MODULE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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